

23. Model Selection and Validation

Tuesday, March 5, 2024

6:36 AM

The issues we want to solve:

- How to set the **hyperparameters** in your algorithm?
(e.g. # rounds T of boosting?)
- How to choose which **algorithm** or hypothesis class $\mathcal{H}$?
- Go beyond theory (w/ unknown constants...) - we want something practical!
- [SS] presents 2 methods, we'll add in classical methods
 - Correct method depends on the data **regime**: lots of data? limited data?

Generic setups

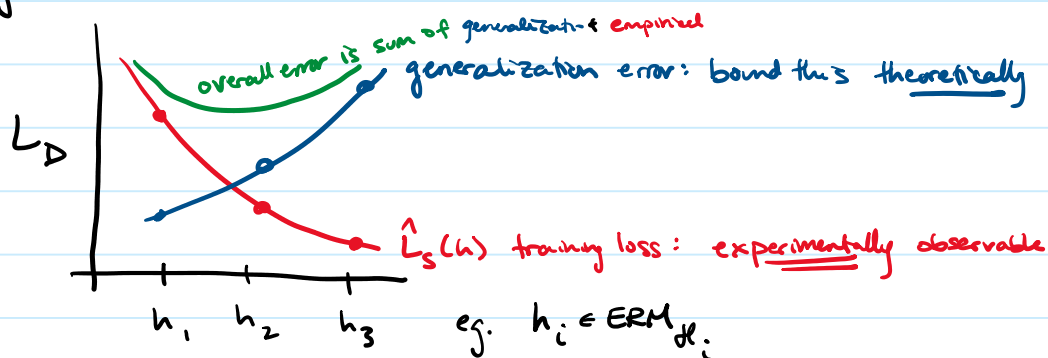
① $\mathcal{H}_1 \subseteq \mathcal{H}_2 \subseteq \mathcal{H}_3 \subseteq \dots$ eg. $\mathcal{H}_n = \{ \text{polynomials of degree } n \text{ or less} \}$. Choose $n \in \mathbb{N}$

② algo is **Tikhonov** regularized least squares aka **Ridge Regression** $\min_w \frac{1}{2} \|X \cdot w - y\|^2 + \alpha \|w\|^2$. Choose $\alpha \geq 0$.

Technique #1: Model Selection via SRM (= Structured Risk Minimization) [§11.1]

mostly of theoretical interest.

Idea:



Similarly:

error decomp.:

$$L_D(h) = \underbrace{\left(L_D(h) - \min_{h' \in \mathcal{H}} L_D(h') \right)}_{\text{estimation error or variance}} + \underbrace{\left(\min_{h' \in \mathcal{H}} L_D(h') \right)}_{\text{approx. error or bias}}$$

SRM based on a (confusing) weight for

obviously this method only works well if theoretical bound captures right behaviour

For theoretical bound, use something like:

• VC dimension

• Minimum Description Length (MDL) (we'll revisit)

• Rademacher Complexity $L_D(h) \leq \hat{L}_S(h) + \underbrace{R_m(\mathcal{H})}_{\text{use in SRM}} + \sqrt{\frac{\log(1/\delta)}{m}}$

23a. Model Selection and Validation

Tuesday, March 5, 2024

1:35 PM

Technique #2: Validation set i.e. train/validation/test data split

Method of choice (simple, rigorous) if data is plentiful

Let $\mathcal{D}$ be any distribution, $S = (z_i)_{i=1}^m$ training set, $h = h_S$,
 $V = (z_i)_{i=1}^{m_v}$ a validation or test set, $V \cap S = \emptyset$, all iid, then

$\forall h$ (either deterministic or independent of V)

$$|\hat{L}_V(h) - L_S(h)| \leq \sqrt{\frac{\log(2/\delta)}{2m_v}} \quad \text{w.p.} \geq 1 - \delta$$

for loss function
w/ range in $[0, 1]$

proof: Hoeffding

$$\text{aka } m_v = \frac{1}{2\epsilon^2} \log(2/\delta)$$

How to use? e.g. model selection:

if we choose among $\{h_1, \dots, h_K\}$ then we have

$$(\text{Simultaneously}) \forall h \in \{h_1, \dots, h_K\} \quad |\hat{L}_V(h) - L_S(h)| \leq \sqrt{\frac{\log(2K/\delta)}{2m_v}}$$

via union bound.

i.e. Select $h \in \arg\min_{i \in [K]} \hat{L}_{V_1}(h_i)$.

$V_1 = \text{validation}$

K too large is a bit
like p-hacking ... hence
always do
test err
also.

↑ like a
"Bonferroni
Correction"

then... re-estimate L_S using $\hat{L}_{V_2}$ $V_2 = \text{testing data}$

to get unbiased estimate (otherwise we're returning an over estimate)

K can be small (ex: $K=2$, comparing two neural net architectures)

or large (ex: $K = \# \text{iterations in SGD}$, i.e. early stopping)
or infinite (ex: $K \in \mathbb{R}$, like in ridge regression)
using a separate validation set is more important

Note: these validation bounds are a posteriori (and fairly tight)

vs.

PAC learning theory bounds are a priori (but loose)

23b. Model Selection and Validation

Tuesday, March 5, 2024

1:56 PM

Techniques not in [SS]

Technique #3 Mallows's C_p / UPRE

Unbiased Predictive Risk Estimate

cf. Vogel's '02 "Comp. Methods for Inverse Problems" and

§ 7.5 in Hastie, Tibshirani, Friedman's "Elements of Stat. Learning"

Model: $y = X \cdot w_{\text{true}} + \eta$ (i.e. agnostic case)

η ← noise

$\begin{matrix} d \\ \boxed{X} \\ m \end{matrix}$ $m > d$

We'll estimate w_{true} via $\hat{w}$

For a given $\hat{w}$, the predictive error is $p = X \cdot (\hat{w} - w_{\text{true}})$

and the predictive risk is $\frac{1}{m} \|p\|^2$ [similar to empirical risk but not quite... no η]

Restrict our attention to estimates $\hat{w}$ that are linear functions of the data y ,

and parameterized by α , i.e. $\hat{w} = w_\alpha := R_\alpha^{-1} y$ (matrix $\neq$ Rademacher complexity!)

Main ex.: ridge regression

$$w_\alpha := \arg \min \frac{1}{2} \|X \cdot w - y\|^2 + \frac{\alpha}{2} \|w\|^2$$

variant: $\|w - w_0\|^2$

always unique sol'n if $\alpha > 0$

$$= (X^T X + \alpha I)^{-1} X^T y$$

normal eq'n (useful theoretically even if we do other tricks numerically)

Question: what value of α to use?

- α too small and we might overfit ($m \gg d$) or no unique sol'n ($m < d$)
- α too large and we bias $\hat{w}$ too much toward 0. (more bias, less variance)

UPRE looks at $\mathbb{E} \frac{1}{m} \|p_\alpha\|^2$,

(r_α, R_α are separate concepts)

$$p_\alpha := X \cdot (w_\alpha - w_{\text{true}})$$

$$r_\alpha = X \cdot w_\alpha - y \quad \text{// residual (so } \frac{1}{m} \|r_\alpha\|^2 \text{ is training loss)}$$

$$= X \cdot w_\alpha - (X w_{\text{true}} + \eta)$$

$$= r_\alpha + \eta$$

observable $\uparrow$ unobservable

$$\text{so } \mathbb{E} \frac{1}{m} \|p_\alpha\|^2 = \frac{1}{m} \mathbb{E} \|r_\alpha\|^2 + \underbrace{\frac{2}{m} \mathbb{E} \langle r_\alpha, \eta \rangle}_{\text{not independent}} + \underbrace{\frac{1}{m} \mathbb{E} \|\eta\|^2}_{= \sigma^2}$$

Assume $\mathbb{E}[\eta] = 0$

$\text{Var}(\eta) = \sigma^2 I_{m \times m}$ ("white" noise)

$$r_\alpha = X w_\alpha - y$$

$$= (X R_\alpha - I) y$$

$$\text{let } P := X R_\alpha \quad (\text{uppercase, } \neq p)$$

$$= (X R_\alpha - I)(X w_{\text{true}} + \eta) = (P - I) \eta + (P - I) \cdot X w_{\text{true}}$$

$$\textcircled{A} = \frac{2}{m} \mathbb{E} \langle (P - I) \eta, \eta \rangle + \frac{2}{m} \mathbb{E} \langle (P - I) X w_{\text{true}}, \eta \rangle = 0 \quad \text{since } \mathbb{E}[\eta] = 0$$

$$= \frac{2}{m} \left(\mathbb{E} \eta^T P \eta - \underbrace{\mathbb{E} \eta^T \eta}_{m \cdot \sigma^2} \right) = \frac{2}{m} \mathbb{E} \eta^T P \eta - 2$$

← η are independent

23c. Model Selection and Validation

Tuesday, March 5, 2024

7:25 PM

what is $\mathbb{E} \eta^T P \eta$? (assuming $\mathbb{E}[\eta] = 0$, $\mathbb{E}[\eta \eta^T] = \text{Var}[\eta] = \sigma^2 \cdot \mathbf{I}$ uncorrelated)

$$\begin{aligned} \mathbb{E} \eta^T P \eta &= \mathbb{E} \text{tr}(\eta^T P \eta) \stackrel{\text{cyclic property}}{=} \mathbb{E} \text{tr}(P \cdot \eta \eta^T) \stackrel{\text{linearity}}{=} \text{tr}(P \mathbb{E} \eta \eta^T) \\ &= \text{tr}(P \sigma^2 \mathbf{I}) = \sigma^2 \text{tr}(P) \\ &= \mathbb{E} \left[\sum_i \eta_i^2 P_{ii} + \sum_{i \neq j} \cancel{\eta_i \eta_j P_{ij}} \right] \stackrel{=0}{=} \mathbb{E} \left[\sum_i \eta_i^2 P_{ii} \right] \end{aligned}$$

2 different ways to see it
trace of a 1x1 matrix
Cyclic property of trace
 $\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB)$

So...

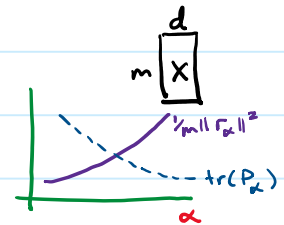
$$\mathbb{E} \frac{1}{n} \|r_\alpha\|^2 = \underbrace{\frac{1}{n} \mathbb{E} \|r_\alpha\|^2}_{\text{a realization is an unbiased estimate.}} + \sigma^2 \left(\frac{2}{n} \text{tr}(P_\alpha) - 1 \right) \quad \text{independent of } \alpha$$

r_α is observable

so the C_p statistic is $U(\alpha) = \underbrace{\frac{1}{n} \|r_\alpha\|^2}_{\text{empirical risk}} + \underbrace{\frac{2\sigma^2}{n} \text{tr}(P_\alpha)}_{\text{penalizes complexity}}$
and UPRE chooses the α that minimizes this statistic.
Hastie calls $\text{tr}(P_\alpha)$ the "effective number of parameters" d

Ex: ridge regression/Tikhonov $P_\alpha = X(X^T X + \alpha \mathbf{I})^{-1} X^T$

$$\begin{aligned} \text{tr}(P_\alpha) &= \text{tr}((X^T X + \alpha \mathbf{I})^{-1} X^T X) \\ &= \text{tr}(\underbrace{(X^T X + \alpha \mathbf{I})^{-1} (X^T X + \alpha \mathbf{I} - \alpha \mathbf{I})}_{=\mathbf{I}}) \\ &= \text{tr}(\mathbf{I}_{d \times d}) - \alpha \text{tr}((X^T X + \alpha \mathbf{I})^{-1}) \\ &= d - \sum_{i=1}^d \frac{\alpha}{\alpha + \lambda_i(X^T X)} \end{aligned}$$



$\lambda_i(X^T X)$ = eigenvalues of $X^T X$

As $\alpha \rightarrow \infty$, $\text{tr}(P_\alpha) = 0$ but empirical risk will be large

$\alpha \rightarrow 0$, $\text{tr}(P_\alpha) = d$ so "effective dimension" = "actual dimension" or "effective number of param."

General version of C_p

$$U = \hat{L}_S(h) + 2 \cdot \frac{d}{n} \hat{\sigma}_\varepsilon^2 \quad \begin{aligned} d &= \text{effective number of parameters} \\ \hat{\sigma}_\varepsilon^2 &= \text{estimate of noise variance} \end{aligned}$$

* High noise $\rightarrow$ use simpler model

23d. Model Selection and Validation (SURE)

Friday, March 13, 2026

9:42 AM

Stein's Unbiased Risk Estimate "SURE" (Stein '81)

generalizes this to nonlinear estimates, $w_\alpha = R_\alpha(y)$ and replaces trace w/ divergence of a Jacobian.

$$\text{i.e. } \text{tr}(P_\alpha) \rightarrow \text{div}(R_\alpha(y)) := \sum_{i=1}^m \frac{\partial r_i}{\partial y_i}$$

Uses **Stein's Lemma** → recently, critical for score matching in diffusion models...
cf. stat.cmu.edu/~larry/=sml/stein.pdf } has typos

Stein's lemma, univariate:

$$Z \sim N(0, 1), f: \mathbb{R} \rightarrow \mathbb{R} \text{ smooth, then } \mathbb{E}[Z \cdot f(Z)] = \mathbb{E}[f'(Z)]$$

proof:

$$\text{let } \phi_\sigma \text{ be pdf of } N(0, \sigma^2), \quad \phi_\sigma(t) = \frac{1}{\sigma\sqrt{2\pi}} e^{-t^2/2\sigma^2}$$

$$\phi'_\sigma(t) = \frac{-t}{\sigma^3\sqrt{2\pi}} e^{-t^2/2\sigma^2}$$

$$\text{so if } \sigma=1, \quad \phi'(t) = -t \cdot \phi(t)$$

Take RHS,

$$\begin{aligned} \mathbb{E}[f'(Z)] &= \int f'(t) \phi(t) dt \stackrel{\text{u.B.P.}}{=} f(t) \phi(t) \Big|_{-\infty}^{\infty} - \int f(t) \underbrace{\phi'(t)}_{-t\phi(t)} dt \\ &= \int t \cdot f(t) \phi(t) dt \\ &= \mathbb{E}[Z \cdot f(Z)] \end{aligned}$$

Multivariate

$$X \sim N(\mu, \sigma^2 I), \mu \in \mathbb{R}^n$$

then

$$\frac{1}{\sigma^2} \mathbb{E}[(X - \mu) f(X)] = \mathbb{E}[\nabla f(X)]$$

↖ transpose? ↗ sum?
if $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is smooth

Connection:

From earlier

$$\mathbb{E} \frac{1}{m} \|P_\alpha\|^2 = \underbrace{\frac{1}{m} \mathbb{E} \|r_\alpha\|^2}_{\text{empirical risk}} + \underbrace{\frac{2}{m} \mathbb{E} \langle r_\alpha, \eta \rangle}_{\text{not independent}} + \underbrace{\frac{1}{m} \mathbb{E} \|\eta\|^2}_{= \sigma^2}$$

Ⓐ
If we assume $\eta \sim N(0, \sigma^2 I)$
then use Stein's lemma...

replace w/ $z^T \cdot P_\alpha$, and in expectation, it's the right thing!